Artificial Intelligence Software, Lecture 4

Model Selection and Sources

MIPT

Center for Cognitive Modeling

Goal of the Lecture

Given a task: «**Find a model for Task X and run it**»

Our goal is to understand what a model is, what type of tasks exists, how to find it, what's important in it and how to launch it.

Architecture vs Weights

Architecture

- Defines the computation graph (layers, connections)
- Example: CNN, Transformer, MLP
- Same architecture can be reused across tasks

Weights

- Learned numerical parameters
- Stored after training
- Define the actual behavior of the model

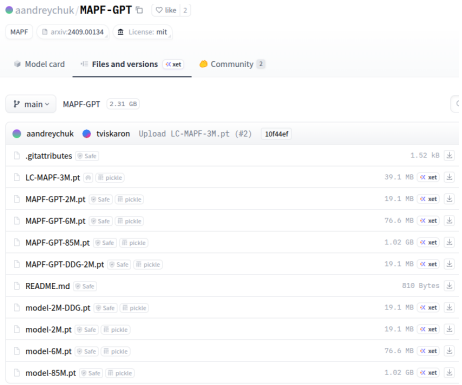
Key idea: Same architecture + different weights = different models

What Does 2M / 85M Parameters Mean?

It’s the number of learnable parameters.

Interpretation:

- 2M parameters ⟶ small model
- 85M parameters ⟶ larger capacity
- More parameters ⟶ more expressive, slower, heavier



How Can Models Share One Architecture?

- Architecture defines the structure (e.g. ResNet18)
- Parameters define the content inside this structure

Same architecture:

- ResNet18 (ImageNet)
- ResNet18 (medical images)

Differences:

- Different weights
- Different datasets
- Different performance

Why Parameter Count Changes

- Depth (number of layers)
- Width (number of neurons / channels)
- Embedding size (Transformers)

Example:

- ResNet18 vs ResNet50
- YOLOv8n vs YOLOv8m vs YOLOv8l

Effect:

- More parameters → better accuracy (usually)
- More memory and compute
- Slower inference

Unified Interface Across Models

- Despite differences, models share a common pattern:

```
output = model(input)
```

Examples:

- Image → classification scores
- Text → sentiment label
- Table → prediction

Important:

- Interface is the same
- Input format is NOT the same

Key Takeaways

- Architecture = structure of computation
- Parameters = learned knowledge
- 2M / 85M = number of parameters

Main insight:

- Same architecture can produce many models
- Difference is in weights and scale

Practical view:

- Models look similar in code
- Real complexity is in data and preprocessing

Types of Tasks

- Computer Vision:
  - classification
  - detection
  - segmentation
- NLP (LLMs, embeddings)
- Tabular (classification/regression)
- Unsupervised (clustering)

Where to Find Models

- Hugging Face — model hub for NLP, CV, multimodal
- GitHub — code and model implementations
- Kaggle — notebooks and pretrained solutions
- Torch Hub — PyTorch pretrained models
- arXiv — research papers (often include models)
- Ultralytics YOLO — ready-to-use CV models for detection and segmentation
- ClearML Model Registry — examples and versioned models for MLOps

Cloud Environments for Running Models

- Hugging Face Spaces — deploy and run NLP, CV, and multimodal models in the cloud without local setup
- Google Colab — free cloud notebooks with CPU/GPU, ready to run Python, PyTorch, TensorFlow
- Kaggle Kernels — notebooks with preloaded datasets and pretrained models
- Ultralytics YOLO HUB — online demos and inference for detection/segmentation
- ClearML Tasks — run training, inference, and manage models remotely with MLOps tools

Model = Code + Weights + Docs

- Code: how to run
- Weights: learned parameters
- Docs: input/output format

How to Choose a Model

- Input format
- Output format
- Metrics
- Hardware requirements
- Latency / size

Preprocessing Matters

- Resize
- Normalize
- Tokenize
- Channel order (RGB/BGR)

Most errors come from wrong input

Examples

all material in extra_materials folder

Example: YOLO Detection

- Input: image
- Output: bounding boxes + labels
- Postprocess: threshold, NMS

Example: NLP Pipeline

- Input: text
- Output: label + score
- Tokenization inside pipeline

Classical ML Still Matters

- Logistic Regression
- Trees / Boosting
- Fast and simple baselines

MLOps: Model Management with ClearML

- Store models — upload trained models to ClearML Model Registry for central access
- Model Details — ready-to-use example projects showing model storage, retrieval, and deployment

Deserve “A” grade!

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